

Voice-Based Concept Understanding Analyser (VBCUA)

Introduction

Project Title

Voice-Based Concept Understanding Analyser (VBCUA)

Team Members

Name	Role
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Project Overview

The Voice-Based Concept Understanding Analyser (VBCUA) is an Artificial Intelligence-based web application developed to evaluate how effectively users understand and explain conceptual topics through spoken communication. Traditional assessments generally focus on written examinations or manual evaluation, making it difficult to measure conceptual understanding, communication skills, and speech quality simultaneously. This project addresses these challenges by integrating speech recognition, semantic similarity analysis, natural language processing, and audio signal processing into a unified platform.

The application enables users to upload or record audio explanations of concepts. It automatically transcribes speech into text, compares the explanation with predefined reference concepts, evaluates semantic

understanding, analyzes speech fluency, and generates comprehensive feedback. The system helps students, educators, interview candidates, trainers, and researchers improve conceptual learning and communication skills through AI-powered automated assessment.

Features

The major features of the Voice-Based Concept Understanding Analyser include:

- Voice recording and audio file upload.
- Automatic speech-to-text conversion using OpenAI Whisper.
- Semantic similarity analysis using Sentence-BERT Transformer models.
- Concept understanding evaluation.
- Speech fluency analysis.
- Filler word detection.
- Pause detection and hesitation analysis.
- RMS energy and speech confidence analysis.
- Waveform visualization.
- AI-generated qualitative feedback.
- Interactive Streamlit dashboard.
- Downloadable PDF evaluation reports.
- Modular and scalable architecture.

Architecture

Frontend

The frontend of the application is developed using Streamlit, a Python-based framework that enables rapid development of interactive web applications. The

interface allows users to upload audio files, view waveform visualizations, analyze speech, and access detailed reports in a clean and responsive environment. Streamlit components provide interactive charts, progress indicators, downloadable reports, and real-time evaluation results.

Backend

The backend is completely implemented using Python. It integrates several AI and signal-processing libraries to perform speech recognition, semantic analysis, scoring, and report generation.

The backend consists of multiple modules including:

- Speech-to-Text Module
- Semantic Similarity Module
- Audio Feature Extraction Module
- Scoring Engine
- Report Generation Module
- Feedback Generation Module

Each module works independently, making the application modular, maintainable, and scalable.

Database

The current version of the application does not require a traditional database. Instead, uploaded audio files, transcripts, waveform images, and generated PDF reports are temporarily stored in local project directories such as:

- audio/
- transcripts/
- reports/
- assets/

Future versions can integrate databases like MongoDB, PostgreSQL, or Firebase to store user history, evaluation records, and authentication details.

Setup Instructions

Prerequisites

Before running the application, ensure the following software is installed:

- Python 3.10 or above
- Git
- Visual Studio Code
- Streamlit
- FFmpeg
- Google Gemini API Key (for AI-generated summaries)
- Internet connection
- Virtual Environment (recommended)

- **Installation**

- **Step 1**

- Clone the GitHub repository.

- `git clone https://github.com/your-repository-link.git`

- **Step 2**

- Navigate to the project folder.

- `cd Voice-Based-Concept-Understanding-Analyser`

- **Step 3**

- Create a virtual environment.

- `python -m venv venv`

- **Step 4**

- Activate the virtual environment.

- **Windows**

- `venv\Scripts\activate`

- **Linux/macOS**

- `source venv/bin/activate`

- **Step 5**

- Install dependencies.

- `pip install -r requirements.txt`

- **Step 6**

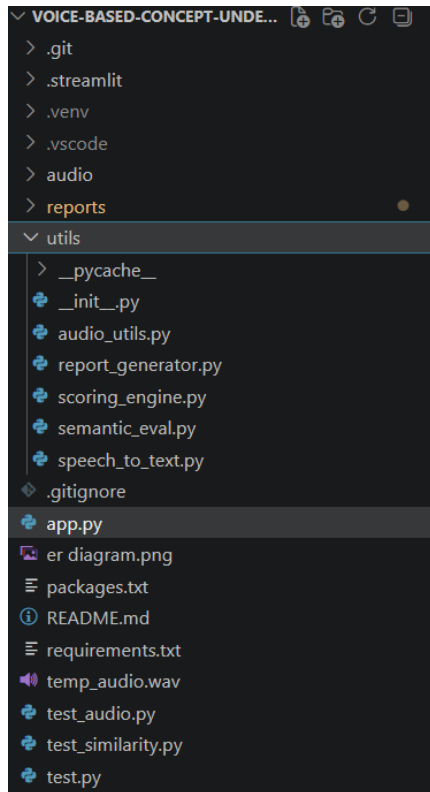
- Configure the Google Gemini API key in the environment variables or configuration file.

- **Step 7**

- Run the application.

- `streamlit run app.py`

Folder Structure



Folder Description

assets/

Stores waveform images, icons, logos, and static resources used by the application.

audio/

Stores uploaded or recorded audio files.

models/

Contains pre-trained AI models used for speech recognition and semantic analysis.

reports/

Stores generated PDF evaluation reports.

transcripts/

Stores speech-to-text transcription files.

utils/

Contains reusable Python modules responsible for speech processing, semantic analysis, feature extraction, scoring, and report generation.

Running the Application

After completing the installation process, activate the virtual environment and execute the following command:

```
streamlit run app.py
```

The application will automatically open in the default web browser.

API Documentation

The current implementation primarily uses internal Python modules instead of REST APIs. However, the application internally performs the following operations:

Module	Description
Speech-to-Text	Converts uploaded speech into text using OpenAI Whisper.
Semantic Similarity	Compares transcribed text with reference concepts using Sentence-BERT embeddings.
Audio Feature Extraction	Extracts pause ratio, RMS energy, filler words, and speech metrics using Librosa.
Scoring Engine	Calculates conceptual understanding scores and speech quality scores.
Report Generator	Creates structured PDF reports with evaluation metrics and AI-generated feedback.

Authentication

The current version of the application does not implement user authentication or authorization mechanisms. All users can access the application locally without logging in.

Future versions may integrate:

- User Registration
- Secure Login

- JWT Authentication
- Role-Based Access Control
- Cloud Database Storage
- User Progress Tracking

User Interface

- The application provides an intuitive and interactive user interface developed using Streamlit. Users can upload or record audio explanations, visualize speech waveforms, view transcribed text, analyze semantic similarity scores, inspect speech fluency metrics, and receive qualitative feedback in real time. The interface also allows users to download detailed PDF reports containing evaluation results, waveform images, AI-generated summaries, and performance metrics.

Homepage:

Voice-Based Concept Understanding Analyser

Automated evaluation of spoken conceptual explanations using AI.

Upload Student Audio (WAV)

Drag and drop file here
Limit 200MB per file • WAV, MP3

Browse files

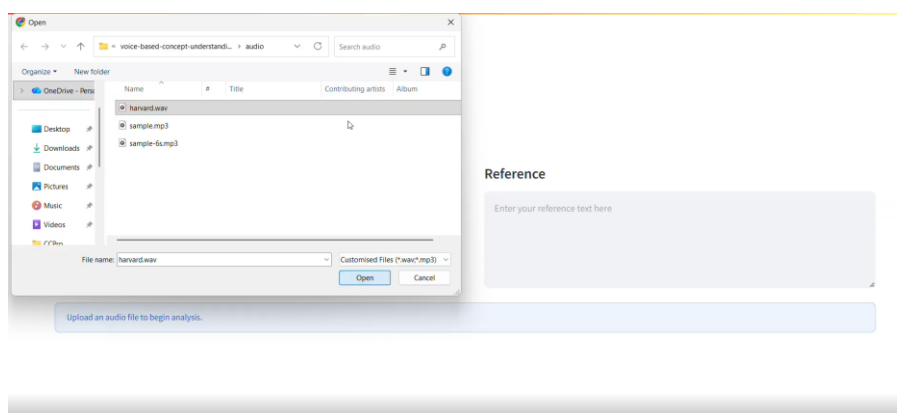
☐ Use Sample Audio (for testing)

Reference

Enter your reference text here

Upload an audio file to begin analysis.

Sample audio:



Reference text insertion:

Voice-Based Concept Understanding Analyser

Automated evaluation of spoken conceptual explanations using AI.

Upload Student Audio (WAV)



Drag and drop file here

Limit 200MB per file • WAV, MP3

Browse files



harvard.wav 3.1MB

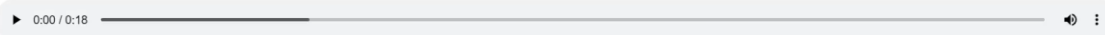


☐ Use Sample Audio (for testing)

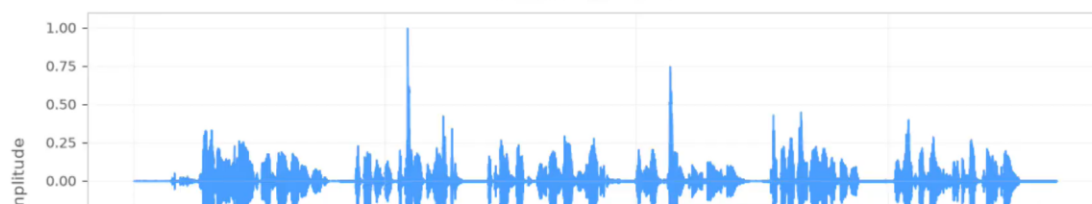
Reference

The stale smell of old beer lingers. It takes heat to bring out the odor. A cold dip restores health and zest. A salt pickle tastes fine with ham. Tacos al pastor are my favorite. A zestful food is the hot cross bun.

Press Ctrl+Enter to apply



Audio Waveform



Waveform:

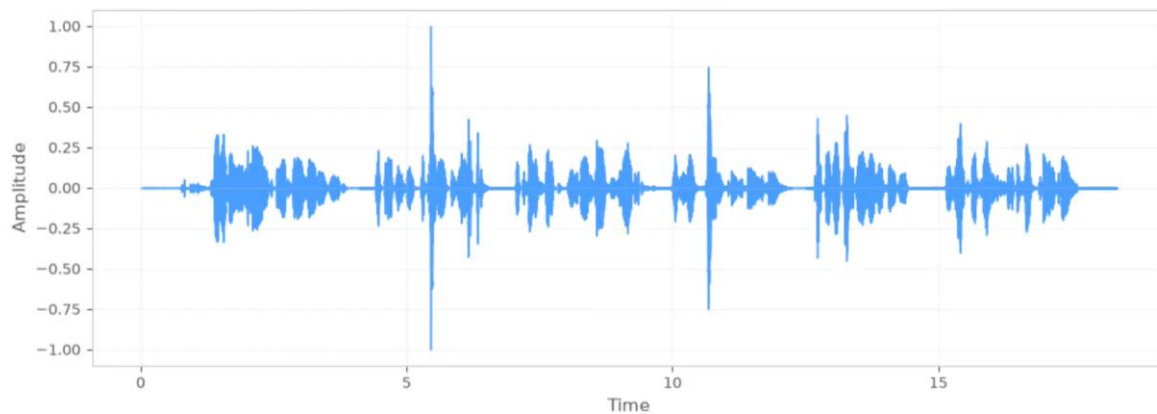


RUNNING...

Stop

Deploy

Audio Waveform



Analyze Concept Understanding

Processing and evaluating...

Analysis of audio:

Deploy

Analyze Concept Understanding

Analysis Completed

Transcribed Explanation

The stale smell of old beer lingers. It takes heat to bring out the odor. A cold dip restores health and zest. A salt pickle tastes fine with ham. Tacos al pastor are my favorite. A zestful food is the hot cross bun.

Final Evaluation

Understanding Score

90/100

Strong Understanding

SEMANTIC SIMILARITY

1.00

FILLER WORD RATIO

0.00

CONFIDENCE (ENERGY)

0.0327

Detailed Score Breakdown & Audio Features

Performance Timing

Download PDF Report

Detailed analysis:

Deploy

pickle tastes fine with ham. Tacos al pastor are my favorite. A zestful food is the hot cross bun.

90/100

Strong Understanding

SEMANTIC SIMILARITY

1.00

FILLER WORD RATIO

0.00

CONFIDENCE (ENERGY)

0.0327

Detailed Score Breakdown & Audio Features

Score Components

Audio Features

Similarity	50/50	Duration (seconds)	18.36
Filler Words	20/20	Average Energy	0.0327
Pause Ratio	5/15	Average Pitch	3243.36
RMS Energy	15/15	Pause Ratio	0.4333
		Zero Crossing Rate	0.0687

Performance Timing

Download PDF Report

Performance testing :

Performance Timing

Audio Loading	0.029s
Transcription (Whisper)	4.058s
Embedding & Similarity	0.129s
Audio Feature Extraction	0.729s
Filler Word Analysis	0.0s
Scoring Engine	0.001s
Total Pipeline	4.946s

Timings measured per-run. Model loading is cached across sessions.

Download PDF Report

Testing

The project has been tested using manual functional testing and module-level validation. Individual modules such as speech transcription, semantic similarity analysis, audio feature extraction, and PDF report generation were verified independently before integration. Test scripts (test.py, test_audio.py, and test_similarity.py) were used to validate the correctness and reliability of different functionalities. Future enhancements may include automated unit testing using PyTest and continuous integration pipelines.

Github link: <https://github.com/dedeepyasriguttikonda-beep/voice-based-concept-understanding-analyser.git>

Future Enhancements

Several enhancements are planned for future versions of the Voice-Based Concept Understanding Analyser. These include support for multiple languages, real-time speech evaluation, user authentication and profile management, cloud-based database integration, personalized learning recommendations, historical performance tracking, advanced emotion and sentiment analysis, interview-specific evaluation modes, integration with Learning Management Systems (LMS), and deployment on cloud platforms such as AWS, Azure, or Google Cloud. Future versions will also explore the use of advanced Large Language Models (LLMs) to provide richer conceptual feedback and adaptive learning suggestions.